RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Monday, 28 September 2026, 3:52 PM
State	Finished
Completed on	Monday, 28 September 2026, 4:06 PM
Time taken	13 mins 8 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int count(int a[], int l, int r, int x)
{
    int c = 0, i;
    for (i = l; i <= r; i++)
        if (a[i] == x)
            c++;
    return c;
}

int majority(int a[], int l, int r)
{
    if (l == r)
        return a[l];

    int mid = (l + r) / 2;

    int left = majority(a, l, mid);
    int right = majority(a, mid + 1, r);

    if (left == right)
        return left;

    int lc = count(a, l, r, left);
    int rc = count(a, l, r, right);
```

```
27
28      if (lc > rc)
29          return left;
30      else
31          return right;
32  }
33
34  int main()
35  {
36      int n, a[50000], i;
37
38      scanf("%d", &n);
39
40      for (i = 0; i < n; i++)
41          scanf("%d", &a[i]);
42
43      printf("%d", majority(a, 0, n - 1));
44
45      return 0;
46  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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